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cheat math sheet



Concepts, Tips, Formulas

Algebra (35% of section):

- **Integer -** A whole number, not a fraction.
- **Domain (x) -** The set of possible x values.
- **Range (y) -** The set of possible y values.
- **x-intercept -** Where $y = 0$.
- **y-intercept -** Where $x = 0$.
- **Standard form:**

$$a_1x + b_1y = c_1$$
$$a_2x + b_2y = c_2$$

One solution	$left_1 \neq left_2$	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
No solution	$left_1 = left_2$ $right_1 \neq right_2$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
∞ solution	$left_1 = left_2$ $right_1 = right_2$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

- **Slope-intercept form:** $y = mx + b$
  - m - slope of the line,
  - b - y-intercept (when $x = 0$),
  - $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$.

One solution	$m_1 \neq m_2$
No solution	$m_1 = m_2$ $b_1 \neq b_2$
∞ solution	$m_1 = m_2$ $b_1 = b_2$

- **Constant/Coefficient:** **X** should be displayed/shown as constant or coefficient = **X** should be visible in the equation/function itself.

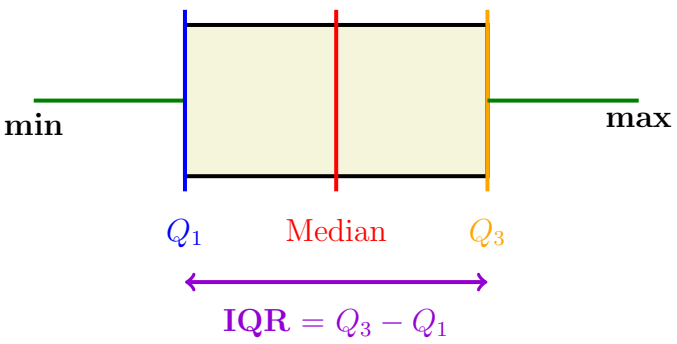
Problem Solving and Data Analysis (15% of section):

- **Mean (Average)** $= \frac{\text{Sum of values}}{\text{Total number of values}}$.
- **Mode -** The most frequently occurring value.
- **Range -** The difference between the highest and lowest values in a dataset. **max-min**
- **Median -** The middle value in a sorted dataset, which divides the data into two equal halves.
- **Low standard deviation:** Data points are close to the mean.
- **High standard deviation:** Data is spread over a wide range.
- **Margin of Error:** How It Changes

  Decreases When:
  - **Bigger Sample Size:** More data reduces uncertainty.
  - **Variability Decreases:** Less spread in the data narrows the error range.

  Increases When:
  - **Smaller Sample Size:** More data increases uncertainty.
  - **Variability Increases:** More spread in the data enlarges the margin.

- **Probability** $= \frac{\text{desired outcomes}}{\text{total possible outcomes}}$.
- **Percent Change**$= \frac{\text{new} - \text{old}}{\text{old}} \cdot 100\%$.
- **Box Plot -** a way of summarizing data by showing:
  - The **minimum value** (left whisker).
  - The **lower quartile** (Q_1), which is the 25% point.
  - The **median** (50% point).
  - The **upper quartile** (Q_3), which is the 75% point.
  - The **maximum value** (right whisker).
  - The **Interquartile Range (IQR)** which is $Q_3 - Q_1$.

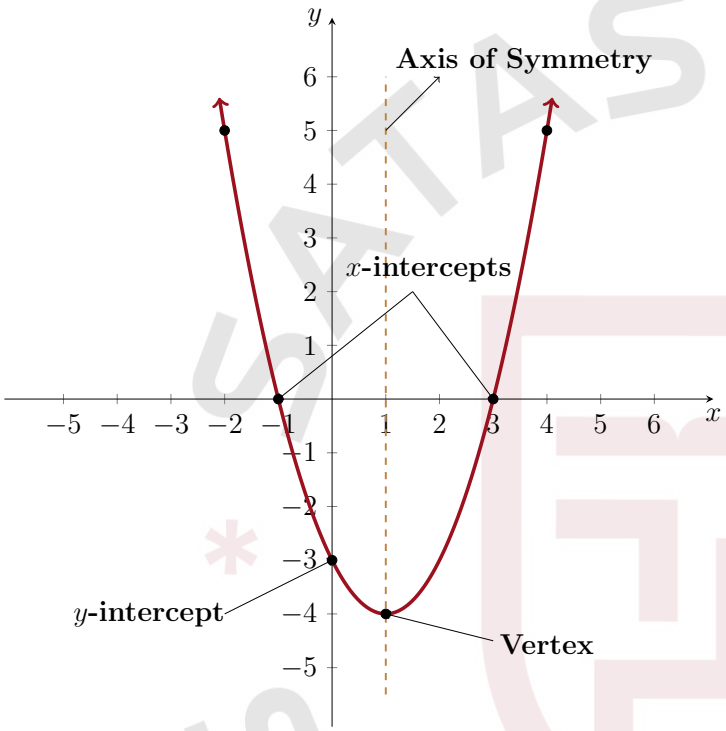


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Advanced Math (35% of section):

Quadratic Functions-Forms

- **Standard Form:** $y = ax^2 + bx + c$, where:
  - c–y-intercept.
- **Factored Form:** $y = a(x-x_1)(x-x_2)$, where:
  - x_1, x_2– x-intercepts.
- **Vertex Form:** $y = a(x-h)^2 + k$, where:
  - (h, k)-vertex coordinate.

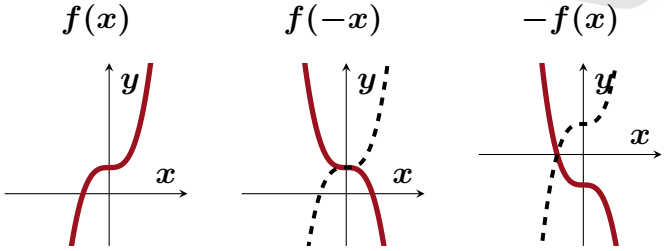


Axis of symmetry and x-coordinate of vertex: $-\dfrac{b}{2a}$

Graph Reflection Rules

- $-f(x) \rightarrow$ Reflection about the **x-axis**
- $f(-x) \rightarrow$ Reflection about the **y-axis**

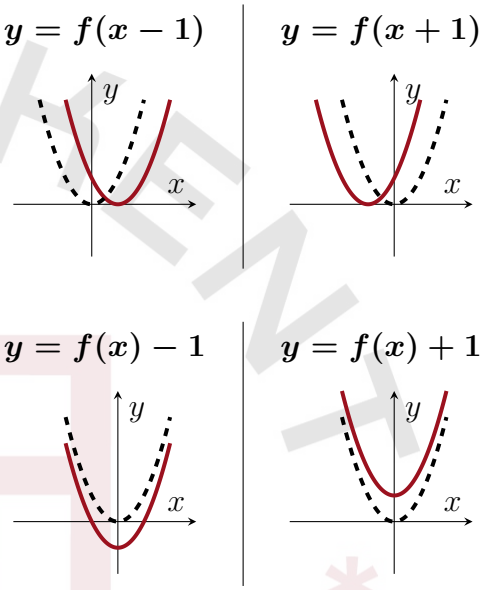
When $f(x) = x^3 + 1$



- **Midpoint** of $A(x_1, y_1), B(x_2, y_2)$:
$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$
- **Distance formula:** $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Function Translation Rules

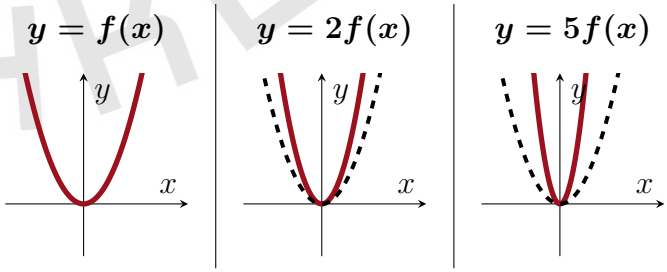
- $f(x+1)$ graph moves **LEFT** one.
- $f(x-1)$ graph moves **RIGHT** one.
- $f(x)+1$ graph moves **UP** one.
- $f(x)-1$ graph moves **DOWN** one.

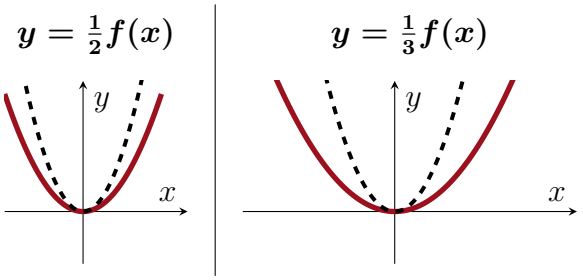


Graph Amplification Rules

x-intercepts remain the same, but for other points:

- $2f(x)$ graph moves **farther** from x-axis.
- $\frac{1}{2}f(x)$ graph moves **closer** to x-axis.



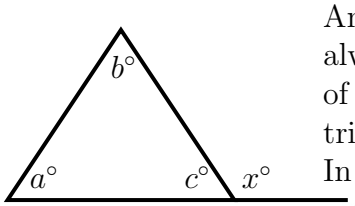


Geometry and Trigonometry (15% of section):

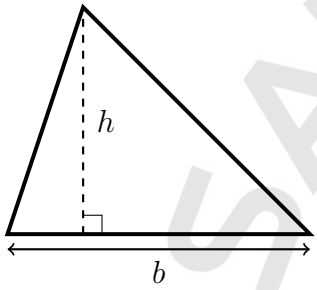
- Sum of interior angles=$180°(n-2)$
- Sum of exterior angles=$360°$

Triangles

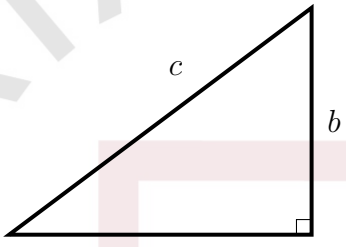
- The sum of the measures in degrees of the angles of a triangle is $180°$.



An **exterior angle** is always equal to the sum of the two angles in the triangle furthest from it. In this case, $x = a + b$.

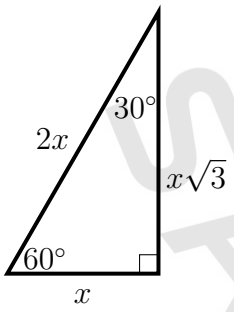


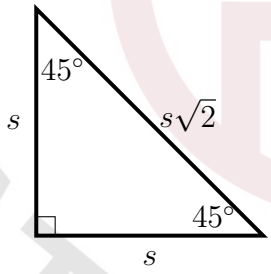
$A = \frac{1}{2}hb$



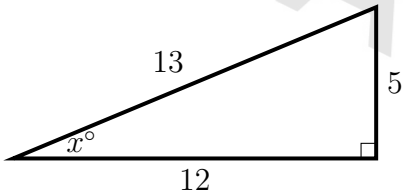
$c^2 = a^2 + b^2$

Special Right Triangles

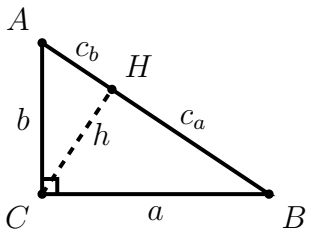




Trigonometry



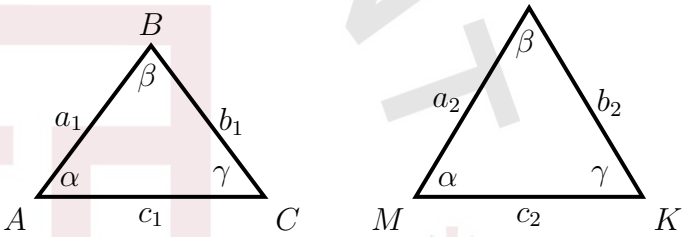
- $\sin x = \dfrac{\text{opposite}}{\text{hypotenuse}} = \dfrac{5}{13}$
- $\cos x = \dfrac{\text{adjacent}}{\text{hypotenuse}} = \dfrac{12}{13}$
- $\tan x = \dfrac{\text{opposite}}{\text{adjacent}} = \dfrac{5}{12}$
- $\sin x = \cos(\dfrac{\pi}{2} - x)$ • $\cos x = \sin(\dfrac{\pi}{2} - x)$
- $\cos x = \sin y$, when $x + y = 90°$



- $h = \sqrt{c_a c_b}$
- $h = \dfrac{ab}{c}$
- $CH-$height$=h$
- $AH = c_b$
- $HB = c_a$

Similarity&Congruency

- Two figures (such as triangles) are **similar** ($\sim$) if they have the same shape, regardless of their size. $\triangle ABC \sim \triangle MNK$

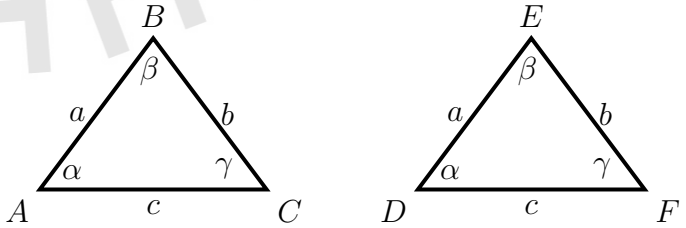


$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{p_1}{p_2} = \frac{h_1}{h_2},$$

$$\left(\frac{a_1}{a_2}\right)^2 = \frac{A_1}{A_2},$$

If two 3D shapes are similar: $\left(\frac{a_1}{a_2}\right)^3 = \frac{V_1}{V_2}$.

- Two figures are **congruent** ($\cong$) if they have exactly the same size and shape. This means that all corresponding sides and angles in the figures are equal.



- **SSS (Side-Side-Side):**
  If all three sides of one triangle are equal to the corresponding three sides of another triangle, then the triangles are congruent.

$$\triangle ABC \cong \triangle DEF$$

When: $AB = DE,\ BC = EF,\ CA = FD.$

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- **SAS (Side-Angle-Side):**
  If two sides and the included angle of one triangle are equal to two sides and the included angle of another triangle, then the triangles are congruent.

  $$\triangle ABC \cong \triangle DEF$$

  When: $AB = DE, \quad \angle B = \angle E, \quad BC = EF.$

- **ASA (Angle-Side-Angle):**
  If two angles and the included side of one triangle are equal to two angles and the included side of another triangle, then the triangles are congruent.

  $$\triangle ABC \cong \triangle DEF$$

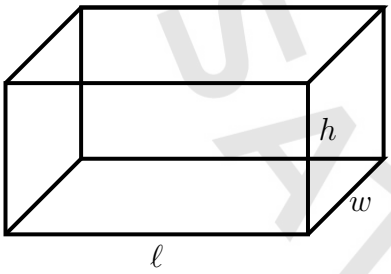
  When: $\angle A = \angle D, \quad AB = DE, \quad \angle B = \angle E.$

- **AAS (Angle-Angle-Side) or SAA (Side-Angle-Angle):**
  If two angles and a non-included side of one triangle are equal to the corresponding two angles and side of another triangle, then the triangles are congruent.

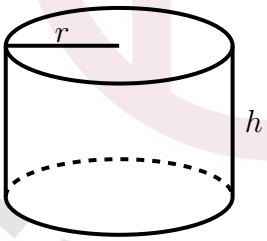
  $$\triangle ABC \cong \triangle DEF$$

  When: $\angle A = \angle D, \quad \angle B = \angle E, \quad BC = EF.$

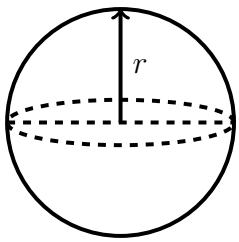
3D shapes



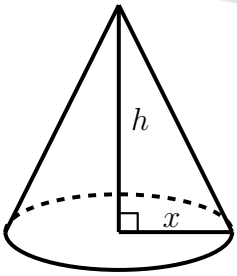
$V = \ell hw$



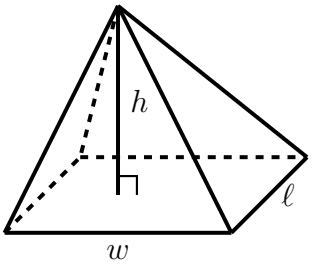
$V = h\pi r^2$



$V = \frac{4}{3}\pi r^3$



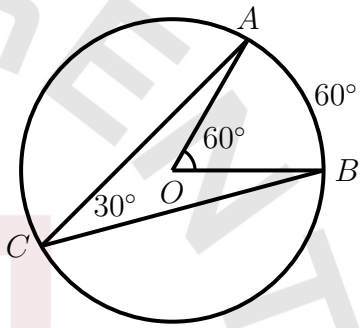
$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}wlh$

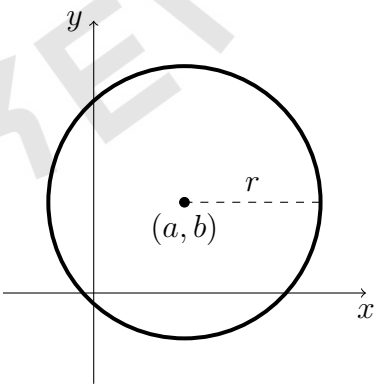
Circle

- The number of degrees of arc in a circle is $360°$.
- The number of radians of arc in a circle is 2π.



- $OA-$radius$=r$
- $\angle AOB-$central angle
- $\angle ACB-$inscribed angle
- **Arc length:** $\dfrac{\theta}{360} \cdot 2\pi r$
- **Area of sector:** $\dfrac{\theta}{360} \cdot \pi r^2$

  $\theta-$ central angle in degrees.



$(x-a)^2 + (y-b)^2 = r^2$

- $r-$radius
- $(a, b)-$center coordinate of circle

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This cheat sheet contains all the essential formulas and key tips you need to tackle SAT Math efficiently. Whether it's algebra, geometry, or advanced topics, we've gathered everything in one place to help you solve questions faster and more accurately. Master the formulas, apply the shortcuts, and boost your SAT Math score!

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